

Addressing the Water Crisis

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2025-05-26

The EU's water crisis by the numbers

Methodology

Results

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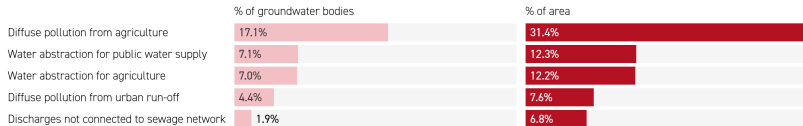
- 40% of the EU's population is already experiencing water scarcity.
- According to the European Environment Agency's freshwater chief Trine Christiansen:

"We simply may not have enough water of good enough quality for the many purposes we would like to use it for."

- The EEA concluded that "**major changes**" to Europe's lifestyles and economic system are the only solution to the continent's deepening water crisis.

Agricultural pollution threatens water quality

Top sources of pressure on the quality of groundwater bodies in the EU, in share of bodies and area affected.



Source: [European Environment Agency](#)

Figure 1: Pressure Sources

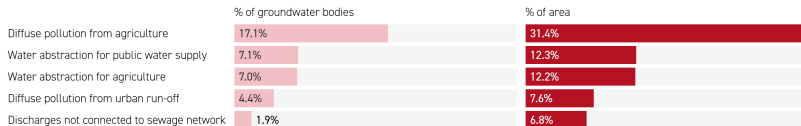
Policy is not enough

- Despite the existence of EU legislation meant to limit the concentration of nitrates in groundwater, progress has been nearly non-existent since 2000. Experts say implementation has been an issue.

Sources of pressure

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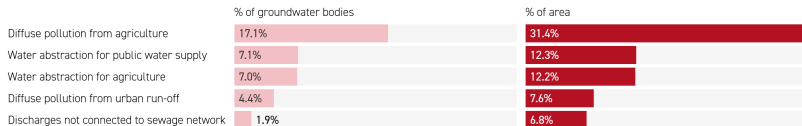
Source: [European Environment Agency](#)

Figure 2: Pressure Sources

Sources of pressure

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Figure 2: Pressure Sources

Nitrate concentration in EU groundwater shows no change

Average concentration of nitrates in EU groundwater, in milligrams of nitrate per liter.

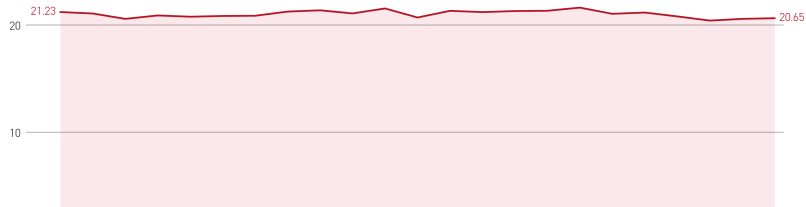




Figure 4: EU's environmental policy-making is complex

- There are currently **52 million people** in the EU + UK living in **water scarcity**.
- In 25 years, up to **65 million people** in the EU + UK could experience **water scarcity**.

- There needs to be a technical component to water management. Cities are perfect places to manage this resource.

Precise water management

- There needs to be a technical component to water management. Cities are perfect places to manage this resource.
- Predicting real-time water use can support more effective and responsive resource management.

Methodology

- This study is based on *Gao et al. (2020)*
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- Essentially they posit a water consumption prediction model with the aid of principal component analysis (PCA) and the integrated deep neural network, which is abbreviated as UWM-Id.

Dataset overview 1997-2024

- x0 – Water Consumption (total liters per person)
- x1 – Resident Population
- x2 – Households with Access to Tap Water ($\times 10$ households)
- x3 – Urbanization Rate (%)
- x4 – Total Number of Employed Persons ($\times 10$)
- x5 – Precipitation (mm)
- x6 – Number of Hospitals and Health Centers
- x7 – Average Relative Humidity (%)
- x8 – Gross Domestic Product ($\times 10$ Euros)
- x9 – Electricity Consumption (GW)

Dataset overview 1997-2024

- x10 – Number of Institutions of Higher Learning (persons)
- x11 – Average Temperature (°C)
- x12 – Natural Gas Consumption ($\times 10$ cubic meters)
- x13 – Engel's Coefficient (%)
- x14 – Irrigated Area (hectares)
- x15 – Water Price if 12 m³ Consumed (€/m³)
- x16 – Average Annual Expenditure by Expenditure Groups (€)
- x17 – Expenditure on Food and Non-Alcoholic Beverages (€)

- All variables were standardized to ensure they had a mean of zero and a standard deviation of one. This is essential for Principal Component Analysis (PCA), which is sensitive to scale.

Initial Matrix of Impact Factors

$$X = \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,m} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{p,1} & x_{p,2} & \cdots & x_{p,m} \end{pmatrix} = \{x_{i,j}\} \quad (1)$$

Step 2: Standardize the Sample Set

$$x_{i,j}^* = \frac{x_{i,j} - \overline{X}_j}{\sigma_j}, \quad (2)$$

$$\overline{X}_j = \frac{\sum_{i=1}^p x_{i,j}}{p}, \quad (3)$$

$$\sigma_j = \sqrt{\frac{\sum_{i=1}^p (x_{i,j} - \overline{X}_j)^2}{p - 1}}, \quad (4)$$

Step 3: Compute the Correlation Matrix

$$R = \frac{1}{p} (X^*)^\top X^* = \begin{bmatrix} r_{1,1} & r_{1,2} & \cdots & r_{1,m} \\ r_{2,1} & r_{2,2} & \cdots & r_{2,m} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p,1} & r_{p,2} & \cdots & r_{p,m} \end{bmatrix}. \quad (5)$$

Step 4: Eigenvalue and Eigenvector Calculation

$$|R - \lambda E| = 0 \quad (\text{Eigenvalue Equation})$$

$$W_j = [w_{1,j}, w_{2,j}, \dots, w_{p,j}], \quad j = 1, 2, \dots, m \quad (6)$$

$$\alpha_i = \frac{\lambda_j}{\sum_{j=1}^m \lambda_j}, \quad \alpha(i) = \sum_{j=1}^m \alpha_j. \quad (7)$$

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$$P = \begin{bmatrix} x_{i,1} & x_{i,2} & \cdots & x_{i,m} \end{bmatrix} \begin{bmatrix} w_{1,1} & w_{1,2} & \cdots & w_{1,m} \\ w_{2,1} & w_{2,2} & \cdots & w_{2,m} \\ \vdots & \vdots & \ddots & \vdots \\ w_{a,1} & w_{a,2} & \cdots & w_{a,m} \end{bmatrix}. \quad (8)$$

- PCA was applied to reduce the dimensionality of our feature space while retaining most of the variance in the data.

Common Architectures for Time Series Prediction

- ARIMA (AutoRegressive Integrated Moving Average)
- Interpretable, great for linear patterns.
- Struggles with non-linearity and multivariate input.
- Feedforward Neural Network (FNN)
- Simple, quick to train.
- No memory of past data.
- Recurrent Neural Networks (RNN)
- Can handle sequential data.
- But forgets long-term patterns.
- GRU (Gated Recurrent Unit)
- Like LSTM but with fewer parameters.
- Slightly less expressive than LSTM in very complex patterns.

Why LSTM (Long Short-Term Memory)

- Remembers Long-Term Patterns
- Handles Multivariate Inputs Naturally
- Proven in Hydrological and Environmental Forecasting
 - Peer-reviewed studies show LSTM can be used to predict water consumption.

LSTM Model Architecture

The LSTM model equations are as follows:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (\text{Forget gate})$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (\text{Input gate})$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C) \quad (\text{Candidate state})$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (\text{Cell state})$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (\text{Output gate})$$

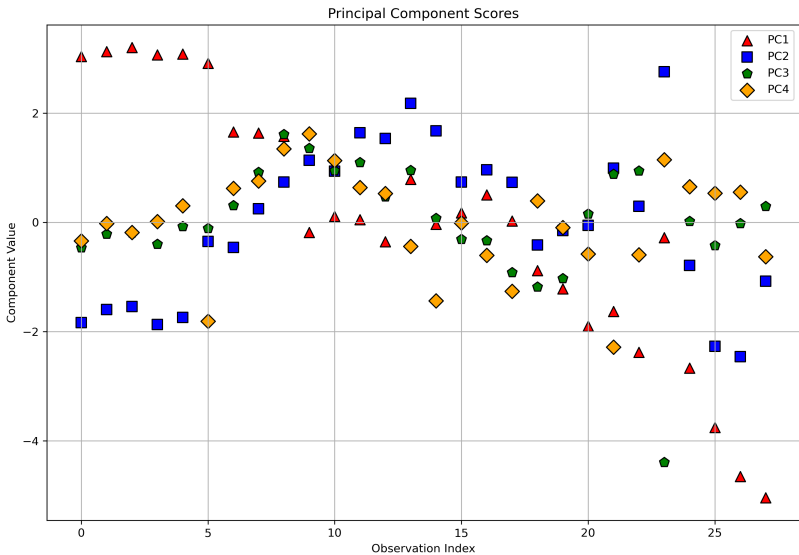
$$h_t = o_t \cdot \tanh(C_t) \quad (\text{Hidden state})$$

Where:

- x_t is the input at time t
- h_t is the hidden (output) state
- C_t is the memory cell state
- σ is the sigmoid function
- $\tanh$ is the hyperbolic tangent

Results

Principal Components



Principal Components

Principal Component	Explained Variance Ratio	Cumulative Variance
PC1	56.9%	56.9%
PC2	20.9%	77.8%
PC3	13.0%	90.8%
PC4	9.2%	100.0%

Table 1: Proportion of Variance Explained by Each Principal Component

Principal Component Interpretation

PC1 – (Most variance explained) Driven by:

- Water Consumption (x0)
- Resident Population (x1)
- GDP (x8)
- Electricity Use (x9)

This component likely captures general urban scale and consumption behavior.

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PC2 Dominated by:

- Precipitation (x5)
- Relative Humidity (x7)
- Average Temperature (x11)

This PC reflects climatic and environmental conditions.

Principal Component Interpretation

PC3 Contributions from:

- Irrigated Area (x14)
- Engel's Coefficient (x13)
- Expenditure on Food (x17)

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PC4 Likely contributors:

- Higher Education (x10)
- Hospitals & Health Centers (x6)
- Gas Consumption (x12)

This PC appears to reflect infrastructure and development indicators.

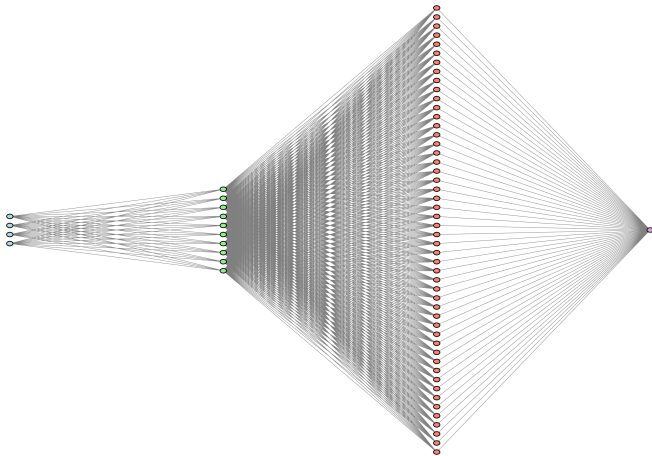


Figure 6: Stylized Model

Model Performance

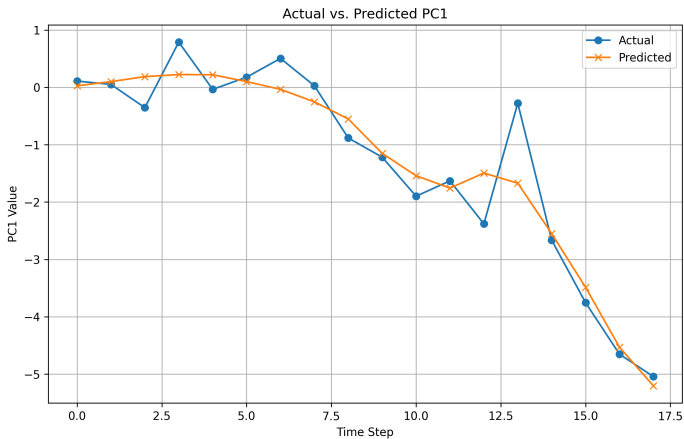


Figure 7: Model Performance

Limitations of what has been done

- Small Dataset (28 samples)
- No Validation or Test Set
- Predicted a Single Output (PC1)
- Assumed Linear Time Dependency (no test of autocorrelation or seasonality)
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